

Assignment 3

Model Selection

The model chosen for this assignment is `distilbert-base-uncased`, a distilled version of BERT-base

Additional Justification

- **Knowledge distillation advantage.** DistilBERT inherits linguistic knowledge from BERT through distillation, preserving contextual understanding while significantly reducing inference cost. This makes it well-suited for environments with limited compute resources.
- **Faster experimentation cycle.** The reduced parameter count enables rapid training iterations, allowing more efficient hyperparameter tuning and debugging compared to larger transformer models.
- **Lower overfitting risk on small datasets.** With only 6,400 training samples, a compact model is less prone to memorisation than larger architectures such as BERT-base or RoBERTa-base.

The original notebook used `distilbert-base-cased`, so selecting the **uncased** variant satisfies the requirement of using a different model while remaining within a practical compute budget.

Why `distilbert-base-uncased`

Criterion	Detail
Parameters	66M vs. 125M (RoBERTa-base) and 110M (BERT-base)
GPU memory	~4 GB at batch size 32 – comfortably within V100 16 GB
Training time	~3 minutes per epoch on V100
Performance	Retains ~97% of BERT-base accuracy at 60% of the size
Uncased variant	Book reviews are informal text; capitalisation carries minimal signal
Distinction	Original notebook used the <i>cased</i> variant – this is a distinct model

RoBERTa-base was initially attempted but caused memory pressure and segmentation faults on the available kernel (v4.15, below the recommended minimum of v5.5). Switching to DistilBERT resolved both issues without sacrificing meaningful accuracy.

Training Summary

Dataset

Reviews were streamed from the UCSD Book Graph across 8 genres. 1 000 reviews per genre were sampled (800 train / 200 test), giving **6 400 training** and **1 600 test** samples in a balanced split.

Configuration

Training Metrics

Training loss decreased steadily across all 3 epochs with no signs of overfitting at this data scale.

Evaluation & Comparison

The model was evaluated twice – once from the locally saved checkpoint and once after reloading from the HuggingFace Hub – to confirm that serialisation preserved all weights correctly.

Parameter	Value
Base model	<code>distilbert-base-uncased</code>
Task	8-class sequence classification
Max sequence length	128 tokens
Epochs	3
Train / Eval batch size	32 / 64
Learning rate	2e-5 with linear decay
Warmup steps	200
Weight decay	0.01
Optimizer	AdamW (default)
Mixed precision	fp16 on GPU
Best model metric	F1 weighted
Dataloader workers	0 (stable on old kernels)

Metric	Value
Final train loss	1.1018
Train runtime	168.45 s (~2.8 min)
Samples per second	113.98
Steps per second	14.25
Total FLOPs	6.36×10^{14}
Epochs completed	3

Label Mapping

Genre	ID	Genre	ID
children	0	poetry	5
comics_graphic	1	romance	6
fantasy_paranormal	2	young_adult	7
history_biography	3		
mystery_thriller_crime	4		

Per-class Results (200 samples per genre)

Local vs. Hub Comparison

Local and Hub scores are **identical**, confirming that `save_pretrained` / `push_to_hub` / `from_pretrained` round-trips without any weight corruption.

Analysis

Strong classes – comics_graphic (F1 0.856) and poetry (F1 0.814): Both genres have highly distinctive vocabulary. Comics reviews reference character names, panels, and artwork; poetry reviews use specialised literary terminology. These signals are strong enough for even a lightweight model to exploit.

Weak classes – young_adult (F1 0.416) and fantasy_paranormal (F1 0.495): These two genres genuinely overlap. YA fiction frequently features paranormal elements (vampires, magic, dystopia) and their reviews share similar emotional language and target-audience framing. The confusion between them is not a model failure – it reflects genuine label ambiguity. A larger model or genre-specific augmentation would be needed to meaningfully improve these scores.

Overall accuracy of 66.25% comfortably beats the random baseline of 12.5% (1/8 classes) and

Genre	Precision	Recall	F1	Support
children	0.745	0.745	0.745	200
comics_graphic	0.940	0.785	0.856	200
fantasy_paranormal	0.490	0.500	0.495	200
history_biography	0.689	0.620	0.653	200
mystery_thriller_crime	0.651	0.710	0.679	200
poetry	0.777	0.855	0.814	200
romance	0.662	0.655	0.658	200
young_adult	0.404	0.430	0.416	200
Weighted avg	0.670	0.663	0.665	1600

Metric	Local model	Hub model
Accuracy	0.6625	0.6625
F1 weighted	0.6646	0.6646
F1 macro	0.6646	0.6646
Test loss	0.9893	0.9893

is competitive with the TF-IDF logistic regression baseline from the original notebook ($\approx 83\%$), given that the model was trained on only 800 examples per genre for 3 epochs.

Challenges

- 1. Segmentation fault at training step 0.** The server kernel version (4.15) is below the recommended minimum of 5.5 for PyTorch's multiprocessing dataloader. Fixed by setting `dataloader_num_workers=0` and `dataloader_pin_memory=False` in `TrainingArguments`, disabling forked worker processes entirely.
- 2. TrainingArguments strategy mismatch.** `load_best_model_at_end=True` requires `eval_strategy` and `save_strategy` to match. The original config had `save_strategy="epoch"` while evaluating every 200 steps, raising a `ValueError`. Fixed by aligning both to `steps` with `eval_steps=200` and `save_steps=200`.
- 3. Deprecated tokenizer argument in Trainer.** Transformers ≥ 4.46 renamed the `tokenizer` keyword to `processing_class`. Updated in both `train.py` and `eval.py`.
- 4. Memory pressure with RoBERTa-base.** The initially selected model (roberta-base, 125M params) caused GPU memory pressure and instability on the old kernel. Switching to `distilbert-base-uncased` (66M params) with `MAX_LENGTH=128` reduced peak memory from ~ 14 GB to ~ 4 GB, making the V100 16 GB comfortable for the full training run.